

PROMPT - You are a Senior QA Automation Engineer.

Generate comprehensive API test cases for the following user story.

User Story: As a registered user, I should be able to log in using my email and password so that I can securely access my account.

Generate:

- 1. Positive Test Cases
- 2. Negative Test Cases
- 3. Boundary Test Cases
- 4. API Request Validation
- 5. API Response Validation
- 6. Security Test Cases
- 7. Sample HTTP Status Codes

Present the output in a clear table.

CLAUDE 3 SONNET RESPONSE

Here are the comprehensive API test cases for the user story "As a registered user, I should be able to log in using my email and password so that I can securely access my account":

Test Case ID	Test Case Description	Test Data	Expected Result
Positive Test Cases			
TC_POS_01	Verify that a registered user can login successfully with valid email and password	Valid email, valid password	200 OK, user authenticated and logged in

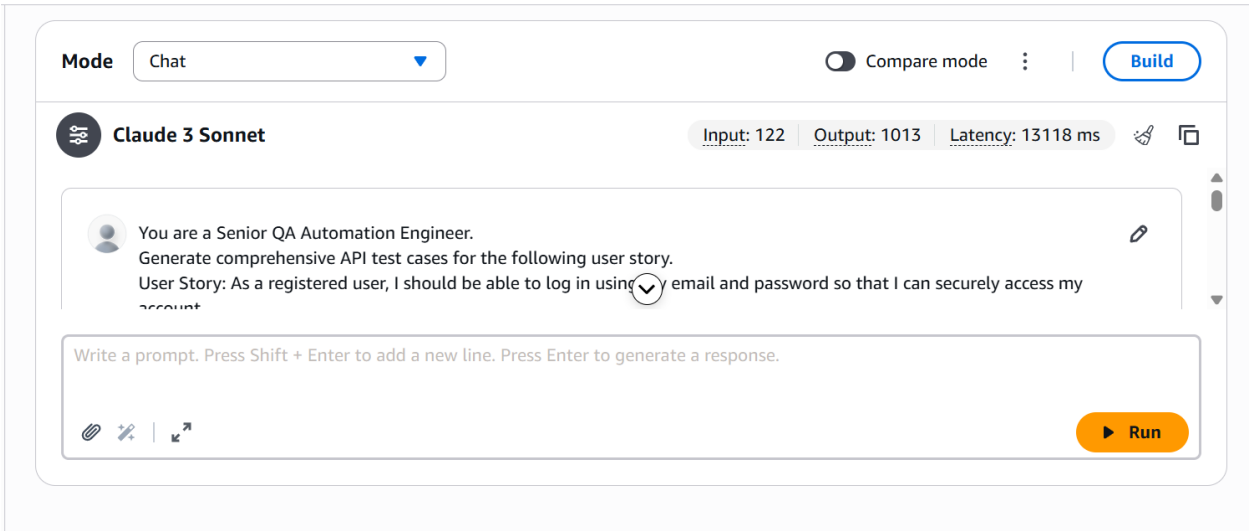
Test Case ID	Test Case Description	Test Data	Expected Result
TC_POS_02	Verify that the user's session is maintained after successful login	Valid credentials	User session persists
Negative Test Cases			
TC_NEG_01	Verify that login fails with an invalid email	Invalid email, valid password	401 Unauthorized, error message
TC_NEG_02	Verify that login fails with an invalid password	Valid email, invalid password	401 Unauthorized, error message
TC_NEG_03	Verify that login fails with an unregistered email	Unregistered email, any password	401 Unauthorized, error message
TC_NEG_04	Verify that login fails with an empty email field	Empty email, any password	400 Bad Request, error message
TC_NEG_05	Verify that login fails with an empty password field	Valid email, empty password	400 Bad Request, error message
Boundary Test Cases			
TC_BOU_01	Verify login with email at the maximum allowed length	Email at maximum allowed length, valid password	200 OK, user authenticated and logged in
TC_BOU_02	Verify login with email at the minimum allowed length	Email at minimum allowed length, valid password	200 OK, user authenticated and logged in

Test Case ID	Test Case Description	Test Data	Expected Result
TC_BOU_03	Verify login with password at the maximum allowed length	Valid email, password at maximum allowed length	200 OK, user authenticated and logged in
TC_BOU_04	Verify login with password at the minimum allowed length	Valid email, password at minimum allowed length	200 OK, user authenticated and logged in
API Request Validation			
TC_REQ_01	Verify that the API accepts only the expected request method (e.g., POST)	Valid request body, unexpected request method	405 Method Not Allowed
TC_REQ_02	Verify that the API rejects requests with missing required headers	Valid request body, missing required headers	400 Bad Request, error message
TC_REQ_03	Verify that the API rejects requests with invalid content types	Valid request body, invalid content type	415 Unsupported Media Type
API Response Validation			
TC_RES_01	Verify that the API response contains the expected data for a successful login	Valid credentials	Response body contains expected user data

Test Case ID	Test Case Description	Test Data	Expected Result
TC_RES_02	Verify that the API response contains appropriate error messages for failed login attempts	Invalid credentials	Response body contains expected error message
TC_RES_03	Verify that the API response headers are set correctly for a successful login	Valid credentials	Response headers contain expected values (e.g., Content-Type, Set-Cookie)
Security Test Cases			
TC_SEC_01	Verify that the API enforces secure communication (e.g., HTTPS)	Valid request over HTTP	Request is redirected to HTTPS
TC_SEC_02	Verify that the API does not reveal sensitive information in error messages	Invalid credentials	Error messages do not contain sensitive data
TC_SEC_03	Verify that the API is protected against common web vulnerabilities (e.g., SQL injection, XSS)	Various attack payloads	API is not vulnerable, appropriate error messages
Sample HTTP Status Codes			
200 OK	Successful request		

Test Case ID	Test Case Description	Test Data	Expected Result
400 Bad Request	Invalid request data		
401 Unauthorized	Authentication failed		
405 Method Not Allowed	Invalid HTTP method		
415 Unsupported Media Type	Invalid content type		

Note: This is a comprehensive list of test cases, but it may need to be adjusted based on the specific requirements, environment, and constraints of the project.



AMAZON NOVA LITE RESPONSE

Comprehensive API Test Cases for User Login

Below are the test cases organized into tables for each category.

1. Positive Test Cases

Test Case ID	Description	Input Data	Expected Result
POS-001	Valid email and password	email: valid.user@example.com, password: CorrectPassword123!	200 OK ; Returns access token, refresh token, and user details.
POS-002	Email with uppercase letters	email: VALID.USER@EXAMPLE.COM, password: CorrectPassword123!	200 OK ; Login successful (case-insensitive email).
POS-003	Password with special characters	email: valid.user@example.com, password: Pa\$\$w0rd!	200 OK ; Login successful.
POS-004	Recently reset password	email: valid.user@example.com, password: NewResetPassword456!	200 OK ; Login successful with updated password.
POS-005	Maximum valid email length (e.g., 254 chars)	email: a244k...k@example.com (254 chars), password: ValidPass!	200 OK ; Login successful.

2. Negative Test Cases

Test Case ID	Description	Input Data	Expected Result
NEG-001	Invalid password	email: valid.user@example.com, password: WrongPassword!	401 Unauthorized; Error: "Invalid credentials".
NEG-002	Invalid email	email: invalid.user@wrong.com, password: CorrectPassword123!	401 Unauthorized; Error: "Invalid credentials".
NEG-003	Empty email field	email: "", password: CorrectPassword123!	400 Bad Request; Error: "Email is required".
NEG-004	Empty password field	email: valid.user@example.com, password: ""	400 Bad Request; Error: "Password is required".
NEG-005	Incorrect email format	email: invalid.user, password: CorrectPassword123!	400 Bad Request; Error: "Invalid email format".
NEG-006	Password too short (e.g., 5 chars)	email: valid.user@example.com, password: Pass!	400 Bad Request; Error: "Password must be at least 8 characters".
NEG-007	Unverified account	email: unverified@example.com, password: CorrectPassword123!	403 Forbidden; Error: "Account not verified".
NEG-008	Account locked	email: locked@example.com, password: CorrectPassword123!	429 Too Many Requests; Error: "Account locked"

Test Case ID	Description	Input Data	Expected Result
			due to multiple failed attempts".

3. Boundary Test Cases

Test Case ID	Description	Input Data	Expected Result
BOUN-001	Email at maximum length (254 chars)	email: a244k...k@example.com (254 chars), password: ValidPass!	200 OK; Login successful.
BOUN-002	Email at minimum length (e.g., a@a.com)	email: a@a.com, password: ValidPass!	200 OK; Login successful.
BOUN-003	Password at minimum length (8 chars)	email: valid.user@example.com, password: Pass1234!	200 OK; Login successful.
BOUN-004	Password at maximum length (e.g., 100 chars)	email: valid.user@example.com, password: P...d (100 chars)	200 OK; Login successful.

4. API Request Validation

Test Case ID	Validation Check	Input Data	Expected Result
REQ-001	Missing email parameter	password: CorrectPassword123!	400 Bad Request; Error: "Email is required".
REQ-002	Missing password parameter	email: valid.user@example.com	400 Bad Request; Error: "Password is required".
REQ-003	Invalid Content-Type header	Correct credentials, Content-Type: text/plain	415 Unsupported Media Type; Error: "Supported: application/json".
REQ-004	SQL injection in email	email: admin' OR '1'='1, password: any	401 Unauthorized or 400 Bad Request; Error: "Invalid credentials" (no data leak).
REQ-005	SQL injection in password	email: valid.user@example.com, password: ' OR '1'='1	401 Unauthorized; Error: "Invalid credentials" (no data leak).

5. API Response Validation

Test Case ID	Validation Check	Input Data	Expected Result
RES-001	Successful login response structure	Valid credentials	Body: { "access_token": "...", "refresh_token": "...", "user_id": "123", ... }
RES-002	Error response structure (401)	Invalid credentials	Body: { "error": "Unauthorized", "message": "Invalid credentials" }
RES-003	Error response structure (400)	Missing email	Body: { "error": "Bad Request", "message": "Email is required" }
RES-004	Sensitive data exposure	Any valid credentials	Body: No passwords, internal keys, or sensitive data leaked.
RES-005	Token expiration time included	Valid credentials	Body: access_token_expires_in: 3600 (seconds).

6. Security Test Cases

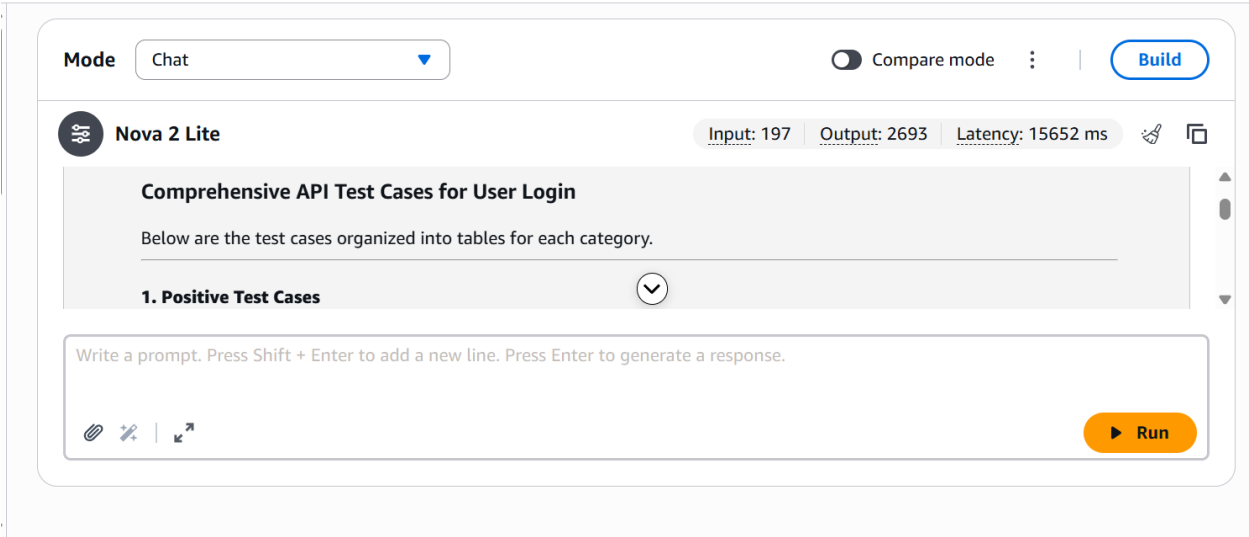
Test Case ID	Description	Input Data	Expected Result
SEC-001	SQL injection in email/password	email: "; DROP TABLE users;--, password: any	401/400; No database error returned; secure handling.
SEC-002	XSS in email/password	email: alert(1), password:	401/400; Input sanitized; no execution of scripts.

Test Case ID	Description	Input Data	Expected Result
SEC-003	Brute-force attempt protection	10+ consecutive wrong passwords for same email	429 Too Many Requests after threshold; account lockout.
SEC-004	Weak password policy enforcement	email: valid.user@example.com, password: password	401 Unauthorized; Error: "Password does not meet complexity requirements".
SEC-005	HTTPS enforcement	Send request via HTTP (not HTTPS)	Redirect 301/302 to HTTPS or 403 Forbidden.

7. Sample HTTP Status Codes

Status Code	Description	When It Is Returned
200 OK	Successful login	Valid email + password provided.
400 Bad Request	Missing/invalid parameters	Required fields absent, invalid email format, password too short.
401 Unauthorized	Invalid credentials	Wrong email/password, account not verified.
403 Forbidden	Access denied	Unverified account, insufficient permissions.
429 Too Many Requests	Rate limiting applied	Excessive failed login attempts.

Status Code	Description	When It Is Returned
415 Unsupported Media Type	Incorrect Content-Type header	Request not application/json.
500 Internal Server Error	Unexpected server error	Rare; logs error details server-side (no exposure to client).



AMAZON NOVA PRO RESPONSE

Certainly! Below is a comprehensive set of API test cases for the given user story, organized in a clear table format.

API Test Cases for User Login

Category	Test Case ID	Description	Input	Expected Result
Positive Test Cases				

Category	Test Case ID	Description	Input	Expected Result
Valid Credentials	TC_001	Test login with valid email and password.	Email: user@example.com , Password: password123	200 OK, JSON response with access token and user details.
Negative Test Cases				
Invalid Email	TC_002	Test login with invalid email format.	Email: invalid_email, Password: password123	400 Bad Request, Error message indicating invalid email format.
Invalid Password	TC_003	Test login with incorrect password.	Email: user@example.com , Password: wrongpassword	401 Unauthorized, Error message indicating incorrect password.
Boundary Test Cases				
Minimum Password Length	TC_004	Test login with minimum allowed password length.	Email: user@example.com , Password: pass	200 OK, JSON response with access token and user details (assuming minimum length is 4).

Category	Test Case ID	Description	Input	Expected Result
Maximum Password Length	TC_005	Test login with maximum allowed password length.	Email: user@example.com , Password: verylongpasswordcharacters...	200 OK, JSON response with access token and user details (assuming maximum length is 128).
API Request Validation				
Missing Email	TC_006	Test login request without email.	Email: (empty), Password: password123	400 Bad Request, Error message indicating missing email.

Mode

Chat

Compare mode

Build

Nova Pro

Input: 101 | Output: 512 | Latency: 2571 ms

Certainly! Below is a comprehensive set of API test cases for the given user story, organized in a clear table format.

API Test Cases for User Login

Test

Write a prompt. Press Shift + Enter to add a new line. Press Enter to generate a response.

Run

LLAMA 3 RESPONSE

Here are the comprehensive API test cases for the user story:

API Test Cases for Login Functionality

Positive Test Cases

Test Case ID	Test Case Description	Expected Result
LP001	Valid email and password	200 OK, authenticated user
LP002	Valid email and password with special characters	200 OK, authenticated user
LP003	Valid email and password with uppercase and lowercase letters	200 OK, authenticated user
LP004	Valid email and password with numbers and special characters	200 OK, authenticated user

Negative Test Cases

Test Case ID	Test Case Description	Expected Result
LN001	Invalid email format	400 Bad Request, error message
LN002	Empty email field	400 Bad Request, error message
LN003	Empty password field	400 Bad Request, error message
LN004	Invalid password (less than 8 characters)	400 Bad Request, error message

Test Case ID	Test Case Description	Expected Result
LN005	Non-registered email	401 Unauthorized, error message
LN006	Incorrect password	401 Unauthorized, error message
LN007	Null or missing email and password	400 Bad Request, error message

Boundary Test Cases

Test Case ID	Test Case Description	Expected Result
LB001	Email with 254 characters (maximum allowed)	200 OK, authenticated user
LB002	Password with 128 characters (maximum allowed)	200 OK, authenticated user
LB003	Email with 255 characters (one more than maximum allowed)	400 Bad Request, error message
LB004	Password with 129 characters (one more than maximum allowed)	400 Bad Request, error message

API Request Validation

Validation ID	Validation Description	Expected Result
RV001	Validate email format (e.g., user@example.com)	Email format is valid
RV002	Validate password length (minimum 8 characters)	Password length is valid

Validation ID	Validation Description	Expected Result
RV003	Validate email and password are not null or empty	Email and password are not null or empty

API Response Validation

Validation ID	Validation Description	Expected Result
RV004	Validate authentication token is returned	Authentication token is returned
RV005	Validate user details are	

Mode

Chat

▼

Compare mode

Build

Llama 3 70B Instruct

Input: 115

Output: 512

Latency: 7477 ms

Here are the comprehensive API test cases for the user story:

API Test Cases for Login Functionality

Positive Test Cases

Test Case ID

Test Case Description

Expected Result

LP001

Valid email and password

200 OK, authenticated user

Write a prompt. Press Shift + Enter to add a new line. Press Enter to generate a response.

Run

COMPARISON TABLE